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30 October 1967
SUPERSEDING
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21 May 1951

MILITARY SPECIFICATION

CUPROUS OXIDE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers reagent grade cuprous oxide.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	-	Box, Fiberboard.
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301	-	Chemicals, Dry and Paste, Packaging and Packing of.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-15011	-	Pallets, Material Handling, Hardwood, Post Construction, 4-Way.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Appearance. The cuprous oxide shall be a reddish-brown powder.

3.2 Chemical characteristics. The cuprous oxide shall conform to the chemical characteristics of table I when tested as specified therein.

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Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as Cu_2O)	96.0	----	4.2.4.1
Insoluble matter in HNO_3	----	0.30	4.2.4.2
Chloride (Cl)	----	0.50	4.2.4.3
Sulfate (SO_4)	----	0.05	4.2.4.4
Substances not precipitated by H_2S	----	0.50	4.2.4.5
Iron (Fe)	----	0.10	4.2.4.6

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Regular production.

4.2.1 Lotting. A lot shall consist of the cuprous oxide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling for test. Sampling shall be conducted in accordance with MIL-STD-105, level S-2. Approximately equal portions of the cuprous oxide shall be taken from each container in the sample, composited, and

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placed in a suitable clean, dry container labeled to identify the lot and containers from which it was taken. The total composited sample shall be no less than 1/4 pound.

4.2.3 Inspection. The composited sample of cuprous oxide shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as Cu_2O). Weigh to the nearest milligram (mg) approximately 0.2 gram (g) of the specimen which has been previously dried to constant weight over sulfuric acid. Mix with 40 milliliters (ml) of 0.1N potassium permanganate solution and 50 ml of 10-percent sulfuric acid and heat to boiling. Titrate the excess permanganate solution with 0.1N oxalic acid solution. Calculate the percent cuprous oxide as follows:

$$\text{Percent cuprous oxide} = \frac{7.15(AB-CD)}{W}$$

where:
A = Milliliters of potassium permanganate solution used,
B = Normality of potassium permanganate solution,
C = Milliliters of oxalic acid solution used,
D = Normality of oxalic acid solution, and
W = Weight of specimen in grams.

4.2.4.2 Insoluble matter in nitric acid. Weigh 5.000 g of the specimen. Heat with 50 ml of 1 to 1 nitric acid until the specimen is dissolved. Dilute with 50 ml of water, filter through a tared filter crucible wieghed to the nearest 0.1 mg (retain the filtrate for 4.2.4.5), wash thoroughly with hot water, dry at 105° ± 5°C for 1 hour, cool, and weigh to the nearest 0.1 mg. Calculate the percent insoluble matter in nitric acid as follows:

$$\text{Percent insoluble matter in nitric acid} = \frac{100(A-B)}{W}$$

where:
A = Weight of filter crucible and insoluble residue in grams,
B = Weight of filter crucible in grams, and
W = Weight of specimen in grams.

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4.2.4.3 Chloride. Weigh to the nearest 0.01 g approximately 1 g of the specimen. Dissolve in 10 ml of 1 to 1 nitric acid and dilute with water. Add an excess of 2-percent silver nitrate solution, shake to coagulate the precipitate and filter through a tared filter crucible weighed to the nearest 0.1 mg. Dry for 1 hour at 105° ± 5°C, cool, and weigh to the nearest 0.1 mg. Calculate the percent chloride as follows:

$$\text{Percent chloride} = \frac{24.7(A-B)}{W}$$

where: A = Weight of filter crucible and precipitate in grams,
B = Weight of filter crucible in grams, and
W = Weight of specimen in grams.

4.2.4.4 Sulfate. Weigh 2.000 g of the specimen, dissolve in 25 ml of dilute aqua regia (1 volume of nitric acid plus 4 volumes of hydrochloric acid plus 6 volumes of water), and evaporate to dryness. Dissolve the residue in 100 ml of 1 to 99 hydrochloric acid and filter. Heat the filtrate to boiling, add 5 ml of 12-percent barium sulfate solution, and allow to stand overnight. Filter through a tared porcelain filter crucible, ignite, cool, and weigh to the nearest 0.1 mg. The weight of the ignited precipitate shall be no more than 0.0024 g greater than the weight obtained in a complete blank test.

4.2.4.5 Substances not precipitated by hydrogen sulfide. To the filtrate from 4.2.4.2, add 20 ml of hydrochloric acid and evaporate to dryness on a steam bath. Take up the residue with 50 ml of water and 0.5 ml of hydrochloric acid, filter, and dilute the filtrate with water to 100 ml. Dilute 40 ml of this solution with 100 ml of water, add 1 ml of sulfuric acid, heat to approximately 70°C, and pass hydrogen sulfide through the solution until the copper is completely precipitated. Cool, dilute to 150 ml with water and filter. Evaporate 75 ml of the filtrate to dryness in a tared porcelain cruciblee weighed to the nearest 0.1 mg, ignite, cool, and weigh to the nearest 0.1 mg. The weight of the ignited residue shall be no more than 0.0050 g greater than the weight obtained in a complete blank test. (Save the residue for 4.2.4.6.)

4.2.4.6 Iron. To the residue from 4.2.4.5, add 1 ml of hydrochloric acid and 0.5 ml of nitric acid. Cover with a watch glass and digest on a steam bath for 15 minutes. Remove the watch glass and evaporate to dryness. Take up the residue with a little warm water, add 1 ml of hydrochloric acid, and dilute to 100 ml with water. To a 10 ml aliquot of the solution, add 2 ml of hydrochloric acid, dilute with 15 ml of water and add 30 to 50 mg of ammonium persulfate crystals and 3 ml of 30-percent ammonium thiocyanate solution. Any red color produced shall be no darker than that produced in a control containing 0.1 mg of iron (Fe) in the same final volume as the specimen and containing the quantities of reagents used in the specimen test.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One-quarter pound ± 0.05 ounce of cuprous oxide shall be packaged in either a type I, class 1 glass bottle utilizing inclosure b (bottles in metal cans) as specified in PPP-C-301 or a class 2, style 2 plastic bottle utilizing closure A or B and outer seal A band or tape as specified in PPP-C-186.

5.1.1.1 Intermediate packaging. Twelve bottles of cuprous oxide, packaged as specified in 5.1.1, shall be intermediately packaged in an upright position in a fiberboard box in a three bottle by four bottle pattern. The approximate inside dimensions of the box shall be 16-1/2 inches by 12-3/8 inches by 6-3/4 inches. Fiberboard pads shall be inserted where needed to assure a tight pack. The box shall conform to style RSC, grade W6c, for a type 2 average load of PPP-B-636. Pads shall conform to type CF, class domestic, variety SW, grade 125 of PPP-F-320. Boxes shall be closed in accordance with requirements for closure by taping class weather-resistant boxes as specified in the appendix to PPP-B-636.

5.1.2 Level C. Quantities of cuprous oxide as specified (see 6.2) shall be packaged to provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Eight boxes of cuprous oxide, intermediately packaged as specified in 5.1.1.1, shall be packed for shipment in a nailed wood box with inside dimensions of 33-3/4 inches by 25-1/2 inches by 14-3/4 inches, conforming to class 2, style 4-1/2, for a type 2 average load of PPP-B-621. The box shall be closed and strapped in accordance with the applicable requirements as specified in the appendix to PPP-B-621.

5.2.2 Level B. Cuprous oxide shall be packed as specified in 5.2.1, except that the box shall conform to either a style RSC, grade V3c box for a type 2 average load of PPP-B-636 or a style optional, class 2 box for a type 2 average load of PPP-B-585 as specified (see 6.2). Boxes shall be closed and strapped as specified in the appendix to the applicable box specification.

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5.2.3 Level C. Cuprous oxide, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to the special marking for unit containers specified in 5.3.1 and any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

5.3.1 Special marking. Each unit container shall be durably and legibly labeled to show the following special information:

(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.
(d) Assay.
(e) Analysis or maximum limits of impurities.
(f) Precautionary marking: "CAUTION - CONTENTS HARMFUL IF SWALLOWED."

5.4 Palletization. When specified in the contract or order, shipping containers of 5.2 shall be palletized in accordance with the applicable requirements of MIL-STD-147 utilizing the hardwood, 4-way, 40 by 48 inch, double-wing pallet of MIL-P-15011 (see 6.2).

6. NOTES

6.1 Intended use. Cuprous oxide is intended for use as a general laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Quantity required for level C packaging.
(d) Type of level B packing box, if required.
(e) Special marking other than 5.3.1, if required.
(f) Palletization, if required.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD
Air Force - 68
DSA-GS

User activity:

Army - SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0399

Coordination Method Code: C

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